

From Privacy to Workflow Integrity: Communication-Graph Metadata in Autonomous Agent Interoperability

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Abstract

Agent-interoperability protocols such as A2A and MCP standardize what agents say to one another, but assume address-based transport over HTTP(S). Such transports protect message *content*, increasingly with end-to-end encryption. What they leave in the clear is the *communication graph*: which agent contacts which, when, and how often. In agent systems this graph is more consequential than a privacy framing suggests. Endpoints are often capability-labeled, workflows are structured and chained, and interactions are coupled to real actions, so an observer of the graph recovers more than a history of past relationships. It can infer the *pending workflow*, the task being assembled and the action likely to follow, and, because these workflows execute at machine speed, act on that inference before the workflow completes. The threat is therefore one of *workflow integrity*, not privacy alone: predictive leverage over autonomous action. We give a threat model for the agent communication graph; identify what makes agent metadata distinctively revealing (semanticity, prospectivity, actuation); define transport- and bootstrap-layer privacy properties (unlinkability, no central observer, deniability, metadata minimization, and discovery privacy) and evaluate candidate transports (SimpleX/SMP, Tor, mixnets) against them; and present an A2A case study in which a metadata-protecting binding is expressible but surfaces the protocol’s implicit identity assumptions. We then test both claims on a generative model of agent workflows anchored to a real A2A capture. From passive metadata alone, with no payloads, a classifier recovers an interaction’s task class well above chance, and does so from only the opening of a workflow. Applied together, the properties drive that recovery sharply back toward chance. Beyond what an observer can *recover*, we measure the leverage of acting on the leak: an adversary that must choose, from a workflow’s opening and under a fixed budget, which workflows to act on realizes in this model most of the advantage a clairvoyant attacker would have over a metadata-blind one, and the same properties that suppress inference suppress this leverage.

1 Introduction

AI agents built by different vendors are increasingly made to interoperate through open protocols. A2A [1], now hosted by the Linux Foundation, and MCP [4] let agents discover one another, delegate tasks, and increasingly transact on behalf of users and organizations. These protocols standardize the *content* and structure of agent messages, but assume a conventional, address-based transport: agents are reachable at URLs or stable names, and messages travel over HTTP(S).

Transport security here has focused, reasonably, on protecting message content: TLS in transit and a growing set of end-to-end schemes that keep payloads from intermediaries. What this leaves untouched is

the *communication graph*: the record of which agent contacts which, when, how often, and how much data flows. Because routing requires addressing and addresses are identifiers, this graph is visible to network observers, relays, and registries even when every payload is encrypted.

The graph is more than a privacy concern. Endpoints are often capability-labeled, workflows are structured and chained, and interactions are coupled to real actions. From such a graph an observer reads the *shape of a task in progress*, not merely a record of past contacts, and at machine speed can act on that shape before the workflow completes. What is exposed is then a matter of *integrity*, not privacy alone: the observer holds predictive leverage over actions that have not yet occurred. Existing agent-protocol

threat models examine authentication, identity, and payload leakage; the communication graph, with its prospective, action-coupled character, has drawn little attention. This paper develops it systematically.

Our contributions are:

1. A **threat model** for the agent-interop *communication graph* as a metadata surface, separate from payload confidentiality (§3, §4).
2. An account of **why agent metadata is distinctively revealing** (its semanticity, prospectivity, and actuation), which reframes the threat from privacy to the *integrity of autonomous workflows* (§5).
3. A **transport- and bootstrap-privacy property framework** (unlinkability, no central observer, deniability, metadata minimization, and discovery privacy) and an evaluation of candidate transports against it (§6, §7).
4. An **A2A case study** showing a metadata-protecting binding is expressible but surfaces the protocol’s implicit identity assumptions, and a **reconciliation** with the ecosystem’s identity and reputation direction (§8, §10).
5. An **empirical evaluation** on simulated agent workflows anchored to a real A2A capture: a label-blind network observer recovers task class well above chance and from only a short prefix of a workflow, and only the full set of properties collapses this recovery toward chance; and, moving from inference to action, that the recovered signal carries decision-theoretic **leverage**: an adversary acting under a budget from a workflow’s opening captures most of the attainable advantage over a blind baseline, which the same properties largely remove (§9).

2 Background

2.1 Agent interoperability protocols

A2A models interoperation as *tasks* exchanged between a client and a remote agent. Agents publish *Agent Cards* (metadata documents at well-known URLs that declare capabilities, endpoints, and authentication) and communicate over one of several *bindings* (JSON-RPC, gRPC, or HTTP+JSON), all over HTTPS [1]. Operations are asynchronous: a call returns immediately, and task updates arrive by polling, server-sent streaming, or push notifications to a client-provided webhook [1, §3.1.7]. A2A also admits *custom protocol bindings* for transports beyond

the core set [1, §5], the extension point we use in §8. MCP plays a complementary (agent-to-tool) role but shares the same address-based, HTTP-oriented assumptions.

2.2 Transport security today

Beyond TLS, recent bindings strengthen content protection: SLIM/SLIMRPC provides broker-less delivery with MLS end-to-end encryption [2, 5], so that no central intermediary reads message content. These mechanisms target confidentiality and, for SLIM, the removal of a trusted broker; none aims at concealing the communication graph.

2.3 Metadata-protecting transports

A separate lineage protects communication *metadata*: mix networks [6], onion routing [8], mixnets [13], and identity-less messaging such as SimpleX’s SMP [15]. These were built for human or general messaging; §7 asks what they offer when repurposed as agent-interop transports.

3 System and Threat Model

3.1 System model

We consider a set of agents $\mathcal{A} = \{a_1, a_2, \dots\}$ that interoperate by exchanging messages under an interop protocol such as A2A or MCP. Two agents communicate over a *transport binding* that realizes the protocol’s abstract operations (request/response, streaming updates, and notifications) over a concrete transport. A binding may route through one or more *intermediaries* (relays, gateways, or brokers), and the protocol may use a *registry* for capability discovery and connection bootstrap.

An *interaction* between a_i and a_j is the set of messages exchanged to complete one logical exchange (in A2A, the lifecycle of a task). Each message m carries a transport-visible descriptor

$$\text{obs}(m) = (\text{src}, \text{dst}, t, \ell, d),$$

its endpoint identifiers, timestamp, length, and direction. Notably $\text{obs}(m)$ excludes the message *content*, which we assume encrypted (Assumption 1).

3.2 The communication graph

Over a period of operation, interactions induce a *communication graph* $G = (V, E)$: V is the set of transport-visible agent identifiers, and each edge

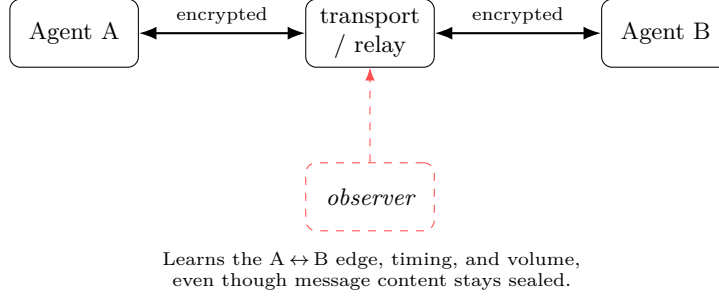


Figure 1: Content encryption protects the payload but not the communication graph. An observer at the network or at an intermediary learns who talks to whom, when, and how often; in agent systems, often capability-labeled endpoints and their sequence further reveal the task in progress (§5).

$e \in E$ records that two endpoints interacted, annotated with timing, frequency, and volume. A *linkage* relation maps transport-visible identifiers to persistent agent or operator identities. The assets we protect are G , this linkage, and, as §5 argues, the predictive leverage over future action that G confers; not the content, which is protected by other means.

3.3 Adversary model

We model *honest-but-curious* adversaries distinguished by vantage point (Table 1); the threat is that G leaks without any active attack. A network observer $\mathcal{N}$ sees $\text{obs}(m)$ for messages on observed links; an intermediary $\mathcal{R}$ sees what it forwards; a registry $\mathcal{G}$ sees discovery lookups and connection bootstrap; a participating or log-retaining endpoint $\mathcal{E}$ sees its own interactions, and more under collusion. Adversaries may collude to widen their coverage of E . Our leakage analysis needs only passive observation: the objective ranges from reconstructing G to inferring the pending workflow it encodes (§5). *Acting* on that inference (to preempt or interfere) may require separate active capabilities, which only strengthen the adversary.

3.4 Trust assumptions and scope

Assumption 1 (Content confidentiality). Message content is end-to-end encrypted under secure primitives; the adversary learns nothing from payloads.

We deliberately grant the *strongest* content protection so as to isolate the metadata axis: any leakage we identify is leakage that content encryption, however strong, does not prevent.

Assumption 2 (No trusted graph custodian). No single party is trusted to observe the full communication graph.

Out of scope are content confidentiality and payload data minimization (assumed handled elsewhere), application-level authorization, and side channels outside the transport.

4 The Communication-Graph Metadata Problem

4.1 Content and metadata are independent

The starting point is a simple but consequential observation: content confidentiality and communication-graph privacy are *independent*, because they protect different things. An interaction can be perfectly content-confidential and still be fully graph-exposed.

Concretely, under Assumption 1 the payload reveals nothing, yet routing still requires the transport to address the destination. When endpoints are named by persistent identifiers, those identifiers appear as *src* and *dst* in every $\text{obs}(m)$ and directly reveal the edge; the timing t , length ℓ , and direction d are visible regardless of encryption. Since G is by definition the set of such edges, any adversary that observes them reconstructs G no matter how strong the content protection is.

We state this as an observation, not a theorem, on purpose: it is close to definitional. The contribution is not the claim that metadata leaks (that is well understood for communication systems in general) but a systematic account of *which* metadata leaks in agent-interop protocols, to *whom* (§3), what a transport must provide to prevent it (§6), and the protocol-level consequences of providing it (§8).

Adversary	Vantage point	Observes
$\mathcal{N}$ network	links / paths	obs(m): endpoints, timing, volume
$\mathcal{R}$ intermediary	relay / gateway / broker	forwarded src/dst, timing, volume
$\mathcal{G}$ registry	discovery service	lookups, connection bootstrap
$\mathcal{E}$ endpoint / log	a participant or its logs	own interactions; more under collusion

Table 1: Adversary classes. All are passive (honest-but-curious) in the base model and may collude to increase coverage of E .

4.2 A walk through the A2A task life-cycle

Let a client a_c delegate a task to a server a_s under A2A, with all content encrypted. At *discovery*, $\mathcal{G}$ observes that a_c resolved a_s . At *connection setup*, $\mathcal{N}$ and any on-path $\mathcal{R}$ observe the $a_c \leftrightarrow a_s$ edge. On *message/send* they observe timing and size; across *streaming or polled updates* they observe cadence and volume; a *push notification* additionally exposes a_c ’s callback endpoint. By *completion*, although no adversary has read a single field of the task, $\mathcal{N}$, $\mathcal{R}$, and $\mathcal{G}$ jointly learn that a_c engaged a_s , when, how often, and how much data flowed, and across many tasks the shape of the agents’ relationships.

4.3 Why current bindings do not address it

The A2A bindings over HTTPS (JSON-RPC, gRPC, HTTP+JSON) protect content with TLS but address agents by URL, so $\mathcal{N}$ and $\mathcal{R}$ obtain the edge directly. SLIM/SLIMRPC removes the central content-reading broker via MLS, yet routes by a persistent structured name; $\mathcal{R}$ and $\mathcal{N}$ still obtain the edge, and the persistent name supplies the linkage of §3. None of these target G : they protect content, which is independent of the graph.

4.4 Problem statement

Definition 1 (Metadata-protecting binding). A transport binding is *metadata-protecting* against an adversary class if, from that adversary’s observations, it cannot reconstruct the communication graph G , nor infer the pending workflow it encodes (§5), beyond a bounded, unlinkable view, as made precise by the properties of §6.

5 Why Agent Metadata Is Different

Generic communication metadata reveals *that* parties communicated. In agent interoperability the same

graph can reveal something stronger, a *task in progress*, because agent interactions differ from generic messaging along three axes.

Semanticity. Agent endpoints, tools, and registry entries are often semantically meaningful rather than opaque addresses. Agent Cards advertise skills, registries are queried by capability, and MCP tools are named by function. Observing that a client contacted a “contract-review” agent or invoked a “payments” tool reveals the *class* of task, not merely that an interaction occurred. This is the explicit-label analogue of website-fingerprinting attacks, where the class of activity is inferred from encrypted-traffic metadata [14], except that here the label is advertised rather than inferred. What an observer actually recovers depends on vantage point and binding: a registry sees capability queries directly, whereas a pure network observer may recover labels only indirectly, through discovery lookups, Agent Card fetches, structured names, or repeated endpoint patterns.

Prospectivity. Agent workflows are structured and chained: discovery precedes delegation, delegation precedes tool invocation, and updates follow. Early steps can therefore predict later ones. An observer who recognizes the opening of a familiar workflow may anticipate its trajectory before it completes, rather than learning of it only afterward.

Actuation. Agent interactions often *trigger actions* directly, without a human reviewing each step. The graph is thus coupled to consequences in the world: influencing or interrupting the observed workflow can change what the agents actually do.

As an illustration, a lookup for a sanctions-screening agent, followed by payment-settlement and contract-review calls, suggests a cross-border transaction being assembled, revealing the *kind* of deal in progress well before it completes, without a single payload being read.

Together these shift the adversary’s objective. From passive observation alone a graph observer may infer

historical relationships and, beyond them, *pending intent and workflow trajectory*. The *harm* comes when the adversary acts on that inference through separate, active channels: poisoning discovery, preempting a negotiation, triggering a competing action. Because the workflow is structured and runs at machine speed, such a move can land before the workflow completes. The pattern is familiar from front-running in decentralized exchanges, where adversaries watch pending-transaction metadata and act ahead of it [7]. Agent interoperability raises the analogous risk wherever workflows are capability-labeled and structured, and potentially across many application domains rather than one.

The framing shifts accordingly. Protecting the communication graph is not merely a privacy question, concealing who interacts; it concerns the *integrity and contestability of autonomous workflows*: their freedom to execute, and to be steered by their principals rather than by an outside observer who holds predictive leverage over machine-speed action. Section 9 measures how much intent the graph leaks, inferring task class from endpoint and sequence metadata; here the threat serves as the design motivation. A scope note: what becomes transport-visible is *inter-agent and inter-tool* coordination, not an agent’s internal, local planning.

6 Privacy Properties for Transport and Bootstrap

The following properties span transport and bootstrap and are protocol-independent; for each we note the adversary capability it removes.

Definition 2 (Unlinkability). An adversary cannot tell whether two observed interactions involve the same agent, nor link a transport-visible identifier to a persistent agent identity. This requires that identifiers not be stable across interactions: each interaction uses a fresh identifier unlinkable to the agent’s others.

Identifier freshness is the mechanism; it denies the edge-linkage of §3 to $\mathcal{N}$ and $\mathcal{R}$, and the persistent-identity linkage to all classes.

Definition 3 (No central observer). No single adversary vantage point observes more than a small fraction of E ; reconstructing G requires collusion among multiple independent parties.

This targets the global view of a network observer $\mathcal{N}$ or a shared intermediary $\mathcal{R}$, and follows Assumption 2.

Definition 4 (Deniability). An interaction leaves no transferable transcript that cryptographically binds a

specific agent to participation; any party can plausibly deny it.

This targets a logging or colluding endpoint $\mathcal{E}$.

Definition 5 (Metadata minimization). The observable descriptors (t, ℓ, d) are reduced (e.g., padded, batched, or mixed) so that timing and volume do not distinguish interactions.

This targets traffic analysis by $\mathcal{N}$ and $\mathcal{R}$ that survives even fresh identifiers.

Definition 6 (Discovery privacy). Capability lookup and connection bootstrap do not reveal the requested capability, the selected peer, or the resulting interaction edge to an untrusted registry or transport intermediary.

This targets the registry $\mathcal{G}$ and the early, pre-interaction leakage that §5 identifies as especially sensitive.

Definitions 2–5 are wire-transport properties; §7 evaluates how far real transports meet them. Discovery privacy (Definition 6) is realized at the bootstrap layer instead, and §8 shows how an identity-less binding can provide it through out-of-band exchange. Together they make a binding metadata-protecting (Definition 1) against the corresponding adversaries; in the terms of §5 they bound an adversary’s predictive leverage by denying the identity, timing, and discovery cues that make workflow inference possible, so they protect the integrity of autonomous workflows, not only privacy.

7 Transport Design Space

No transport was designed for agent interoperability; each was built for human or general messaging, so applying it inherits both its protections and its limitations. Table 2 rates candidate transports against the properties of §6, alongside the HTTP(S) and SLIM bindings as baselines. Ratings are qualitative (*strong / partial / weak*); the point is the shape of the trade-off, not a score.

HTTP(S) bindings (incl. SLIM). Agents are addressed by persistent URL or structured name, so unlinkability is weak and a network observer or intermediary obtains the edge directly. SLIM removes the central content-reading broker (an improvement over a single shared intermediary) but still routes by persistent name and provides no identifier freshness, mixing, or deniability.

Transport	Unlinkability	No central obs.	Deniability	Metadata min.	Cost
HTTP(S) / current bindings	weak	weak	weak	weak	low latency, mature
SLIM / SLIMRPC	weak	partial	weak	weak	low latency, maturing
SimpleX / SMP	strong	strong	strong	partial	async, modest throughput
Tor onion services	weak	strong	partial	partial	moderate latency, mature
Mixnet (e.g. Nym)	strong	strong	strong	strong	high latency, emerging

Table 2: Candidate transports rated against the four wire-transport properties of §6; discovery privacy is a bootstrap-layer concern, addressed in §8. Ratings are qualitative; the point is the trade-off, not a score.

SimpleX / SMP. Identity-less by construction: connections are bootstrapped out of band and carried over unidirectional queues with per-queue identifiers and no global account, giving strong unlinkability and, with separate and rotating relays, no single observer of the graph; deniability is a design goal. Its weak point is metadata minimization (a relay still sees the timing and volume of the queues it hosts), so traffic-analysis defenses are only partial. The model is asynchronous with modest throughput.

Tor onion services. Strong at hiding network location and distributing trust across relays, but a published onion address is a *persistent* identifier, so unlinkability is weak when agents reuse addresses, and a global passive adversary can mount traffic correlation. Maturity is high, latency moderate.

Mixnets (e.g. Nym). Purpose-built for metadata protection: per-packet unlinkable formats, distributed mixing, and cover traffic earn strong ratings on unlinkability, no-central-observer, and metadata minimization. The cost is high latency and lower maturity, precisely the trade-off a deployment must weigh.

Takeaway. No transport provides all four wire-transport properties cheaply; they trace a privacy/latency frontier. SMP is a strong first instantiation because it is identity-less and ships today; a mixnet is stronger on traffic analysis at a latency cost; Tor is the most mature but weakest on unlinkability. The properties of §6, not any single transport, are the portable target.

8 Case Study: A Metadata-Protecting Binding for A2A

A2A is a useful case study because it already admits transports beyond its core set through *custom protocol bindings*, and because its operations are already asynchronous: an operation returns immediately and updates arrive by polling, streaming, or push. A

metadata-protecting binding is therefore expressible in principle. The instructive result is what one meets in trying: mapping A2A onto an identity-less transport (we use SMP) surfaces three *implicit identity assumptions* that the specification never states because, over HTTP, they always hold.

Assumption 1: push notifications assume an HTTP-reachable client. A2A push delivers task updates to a client-provided webhook URL [1, §3.1.7], presuming the client has a stable, reachable address; an identity-less client has none. This is *surmountable*: server-initiated delivery re-maps onto the transport’s own asynchronous channel. At task creation the client supplies a reply queue it controls, and the server posts updates there. Because A2A already treats push as one of several interchangeable update mechanisms, the semantics are preserved; only the carrier changes.

Assumption 2: authentication is identity-based. A2A authentication is declared in the Agent Card and is identity-bearing (mutual TLS [1, §4.5.6], OAuth/OIDC, keys tied to a principal). An identity-less transport cannot present a stable principal. This is the *genuine mismatch*: schemes that require a verifiable persistent identity (a client certificate, an OIDC subject) do not translate. What does translate is a different trust basis: channel binding from the out-of-band handshake, plus capability- or credential-based authorization via selectively disclosed attestations (§10). This establishes *what* a peer is entitled to without fixing *who* it persistently is.

Assumption 3: discovery assumes addressable endpoints. A2A discovery resolves an Agent Card at a well-known URL [1] and selects a binding from its declared endpoint. With no addressable endpoint, both the card’s location and the endpoint it advertises must change form. This is *surmountable* but needs a different bootstrap: capabilities are exchanged out of band (an invitation carries or precedes the Agent Card), and the card declares a rendezvous or invitation mechanism in place of a URL. The capability *content*

of the card is unaffected; its addressing model is what gives way.

What the case study shows. Two of the three assumptions (push, discovery) are surmountable by re-mapping onto asynchronous and out-of-band mechanisms the transport already provides; one (identity-based authentication) is a semantic mismatch that forces a different, credential-based trust basis. None is stated in the specification, because over an address-based transport they hold for free. Naming them is useful independently of whether an identity-less binding is ever standardized: they delimit exactly where interoperability and persistent identity are entangled. They are also where an adversary would act: discovery and push are the early, action-coupled steps whose visibility enables the preemption of §5.

9 Empirical Evaluation

Two claims from the earlier sections invite a test. The threat model holds that graph metadata leaks *pending workflow intent* (§5); the property framework holds that a defined *set* of properties removes it (§6). We evaluate both, and then ask the third, decision-theoretic question the integrity framing demands: what the recovered signal is *worth* to an adversary that acts on it (§9.4). The aim is a controlled demonstration rather than a field measurement: that intent is recoverable from passive metadata, that it is recoverable *early*, and that the properties reduce recovery sharply toward chance, and only in combination.¹

9.1 Setup

No public corpus of agent-interop traces exists, so we sample workflows from a generative model of the A2A task lifecycle: discovery, delegation, streamed updates, completion. Each *task class* is a stochastic process over capability-typed stages, drawn from a shared vocabulary with tunable overlap, so the *set* of capabilities alone does not identify the class. Several agents serve each capability and some are multi-skill, so a transport-visible identifier is *not* merely a labeled capability. Timing and size profiles attach to capabilities rather than classes and are shared across them; all class signal therefore flows through *which* capabilities are invoked and *in what order*. Each message yields the descriptor $\text{obs}(m)$ of §3, and content is

never observed. To check realism, we anchor the generator to a real capture from the A2A reference SDK, a live task lifecycle driven over HTTP, and confirm that it matches the lifecycle’s structure and scale.

An *adversary view* projects a trace onto what a single vantage point sees (Table 1). The *registry* view sees the semantic capability labels named in discovery queries. The *network* view sees only opaque endpoint identifiers, timing, volume, and direction, and no semantic labels. A classifier then predicts the latent task class. Following the website-fingerprinting tradition [14], we treat its accuracy above chance as a conservative indicator of leakage: an unoptimized decoder can only understate it, so we deliberately leave it untuned. Chance is $1/K$ for K balanced classes, and we report cross-validated accuracy with a bootstrap 95% confidence interval.

9.2 Leakage is recoverable in the model, and prospective

With $K = 8$ balanced classes, chance is 0.125. The registry view recovers the task class at 1.00, near-tautologically, since it observes the labels directly. The result that matters is the *label-blind* network view, which recovers the class at 0.99 (Fig. 2). An observer of pure transport metadata is thus nearly as informed as one reading the advertised labels: persistent identifiers, together with capability-correlated timing and volume, let it reconstruct the capability footprint indirectly. This is the semanticity of §5 made concrete.

Recovery is also *prospective*. Given only the first tenth of a workflow, the network view already predicts its class at 0.70, roughly $5.6\times$ chance, and accuracy climbs toward certainty as more of the workflow is observed (Fig. 3). The opening of a workflow predicts its trajectory, exactly the predictive leverage §5 singles out as distinctive to agent metadata.

9.3 The properties neutralize the leak, but only as a set

We next realize the properties of §6 as transforms on the observed traffic and re-measure (Fig. 4). On its own, each wire property barely moves the network observer. *Unlinkability* (fresh per-interaction identifiers) closes the persistent-identifier channel but leaves the timing and volume fingerprint, holding accuracy at 0.95. *Metadata minimization* (padding and a batched cadence) closes timing and volume but leaves the identifiers, holding it at 0.99. Only the two together collapse recovery, to 0.42. The registry view is a separate channel: neither wire property touches it, and it

¹Code and data: <https://github.com/dangoldbj/agent-metadata-privacy>.

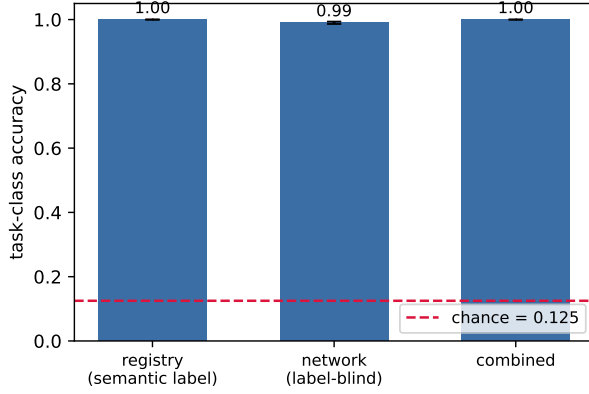


Figure 2: Task class recovered from communication-graph metadata, by adversary view ($K = 8$, chance 0.125; error bars are bootstrap 95% CIs). Even the label-blind *network* view, seeing only $\text{obs}(m)$, recovers the class far above chance.

falls only to *discovery privacy* ($1.00 \rightarrow 0.125$, exactly chance). The threat therefore yields only to the full *set* of properties, each matched to a channel; partial measures do not suffice, a content-protecting binding that retains persistent names being one such case (§4). The residual 0.42 stays above chance because a *structural* channel remains: the message counts and sequence shape that the wire properties do not target and that cover traffic would address (§7).

9.4 Actuation: the value of acting on the leak

Leakage and prospectivity are properties of recoverability: how much the metadata tells an observer about the task. Whether that knowledge bears on workflow *integrity* is a separate, decision-theoretic question, since an observer that cannot act on what it learns poses no integrity threat. We therefore model an adversary that must *act* under a budget and measure what the metadata is worth to it.

Definition 7 (Actuation game and value of metadata). Among N concurrent workflows, each w carries an adversary value $v(w) \geq 0$. By a decision deadline f , having observed only the leading fraction f of every workflow, a metadata-only adversary commits a budget of B interventions, choosing a set S with $|S| = B$ to maximize $J(S) = \sum_{w \in S} v(w)$, and ranks workflows using the label-blind network view of the observed prefix alone. With J_{inf} , J_{blind} , and J_{orc} the objective under the metadata-informed, uniformly random, and true-value selections, the *value of metadata* is $\text{VoM}(B, f) = J_{\text{inf}} - J_{\text{blind}}$ and the *capture ratio* is

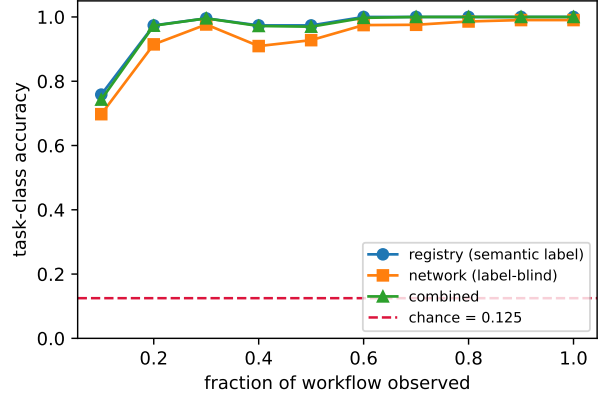


Figure 3: Prospectivity: accuracy as a function of the fraction of the workflow observed. From only its opening, the network view predicts the pending task class well above chance.

$\kappa = (J_{\text{inf}} - J_{\text{blind}}) / (J_{\text{orc}} - J_{\text{blind}})$, normalized so that 0 is the blind baseline and 1 the oracle; a ranking worse than random can fall below 0, and in our experiments $\kappa \in [0, 1]$. It is the share of the attainable advantage the adversary realizes from metadata alone.

We instantiate the game minimally, adding nothing to the generator. One task class is the adversary’s target, $v(w) = 1$ if w is of that class and 0 otherwise; the budget equals one class’s mass; the ranking is the network observer’s out-of-fold posterior on the target, from exactly the prefix features used in the preceding subsections. Then J_{inf} is the count of true target workflows among the top- B by that posterior, and J_{blind} and J_{orc} are closed-form; we average over the choice of target. What VoM and κ capture is *selection leverage*: the advantage of picking the right workflows to act on, from the opening alone. That is the precondition for changing outcomes, not proof that they change; the latter would mean acting against a live binding (§10).

The value of metadata is substantial and, like the inference beneath it, prospective. Deciding from only the opening fifth of each workflow, the adversary captures $\kappa \approx 0.90$ of the attainable advantage over the blind baseline (Fig. 5); from only the first tenth it already captures about two-thirds, climbing toward unity as more of the workflow is seen (Fig. 6, left). Under a budget, knowing the task early is most of the way to acting on it.

Actuation is not a restatement of leakage. The value of metadata is the product of two independent factors (an early-decidable signal and a budget to spend), and collapses if either is absent: with no budget there is nothing to actuate, so $\text{VoM} \rightarrow 0$ as

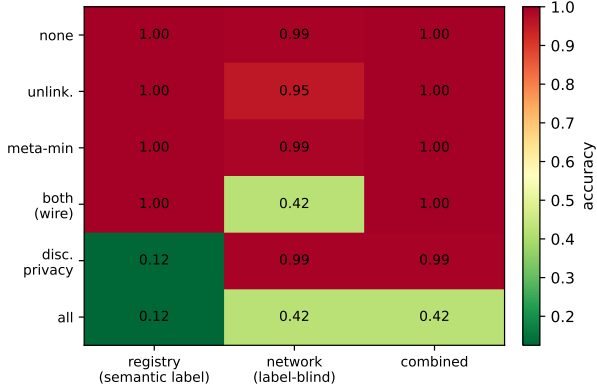


Figure 4: Accuracy under each property (rows) for each adversary view (columns); red is leaking, green is protected. The network observer falls only when unlinkability and metadata minimization are combined (“both”); the registry observer falls only to discovery privacy. No single property suffices.

the budget shrinks (Fig. 6, right), and with no early signal the ranking is uninformative and $\kappa \rightarrow 0$. The axis is genuinely separate from recoverability, and website-fingerprinting, which bounds the signal factor alone, does not speak to it.

The defense carries over, and again only as a set. Each wire property alone leaves the leverage near its unprotected level ($\kappa = 0.83$ under unlinkability, 0.92 under metadata minimization); the two together drive it to 0.12, essentially the blind baseline (Fig. 5). Notably the leverage falls further than inference itself: under both properties the label-blind observer still recovers task class at 0.42 (Fig. 4), yet that residual labeling power buys almost no leverage, because selecting the highest-value workflows under a tight budget demands a precision the residual channel lacks. Discovery privacy, which does not touch the network view, leaves κ unchanged, as expected. Selection leverage is downstream of inference: with recovery gone, there is nothing left to target. The integrity defense thus follows from the privacy defense, since the properties that suppress what an observer can recover suppress the leverage it gains.

9.5 Robustness and scope

The effect is structural rather than tuned. Across sweeps of the number of classes, the capability overlap, and the timing noise, the network view stays 4–14 $\times$ above chance, the short-prefix prediction stays above chance, and the two wire properties always collapse recovery. We therefore claim the *structure* of the effect, not its precise magnitude; this holds for the

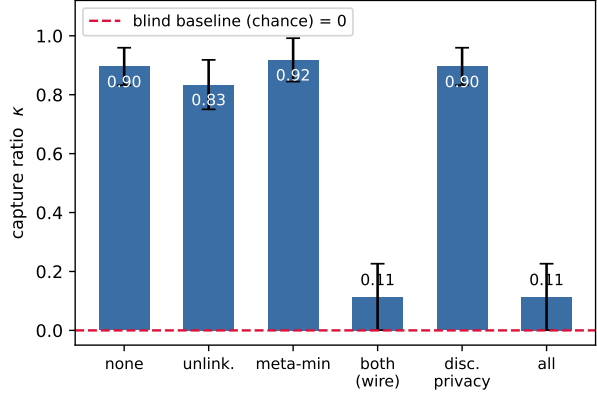


Figure 5: Actuation. Capture ratio κ by privacy property, at an early decision deadline ($f = 0.2$) and a budget equal to one task class’s mass, averaged over targets (error bars span ± 1.96 standard errors across target classes; chance, the blind baseline, is 0). The integrity analogue of Fig. 4: only the combined wire properties (“both”) collapse the leverage, and discovery privacy, which the label-blind observer ignores, does not.

value of metadata too. Two limitations are intrinsic. The workflows are simulated and calibrated to a small real capture rather than a labeled real corpus, so the magnitude is generator-dependent. And the actuation result (§9.4) measures leverage *in the model*: it quantifies the advantage a budgeted adversary’s selection gains from metadata over the simulated workflow population, the decision-theoretic core of the integrity threat, but stops short of a live exploit against a deployed binding. The evaluation thus substantiates all three axes (semanticity, prospectivity, and actuation) and the efficacy of the properties against each; demonstrating end-to-end manipulation on real agent traffic is the natural next step.

10 Discussion

10.1 Reputation and trust without a global graph

A natural objection, especially in the current agent ecosystem, is that trust and reputation require persistent identity and an observable history of interactions, which metadata privacy appears to break. The tension is real but narrower than it first appears. It conflicts only with *global-observation* reputation: a registry or ledger that watches all interactions to compute scores. That design is fundamentally incompatible with unlinkability, and is itself a graph-

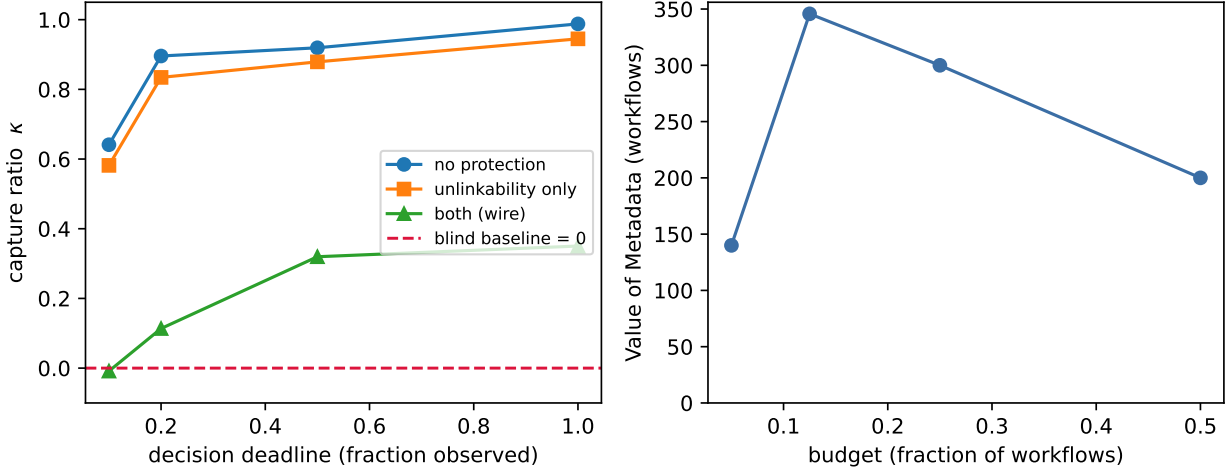


Figure 6: Actuation is the product of inference and budget, vanishing on either edge. Left: capture ratio against the decision deadline; leverage tracks prospectivity, is substantial even from a short prefix, and the combined wire properties hold it at the blind baseline. Right: value of metadata against budget (no protection, full workflow); it vanishes without a budget to spend and peaks where the budget is scarce relative to the target set, the signature of a value-of-information quantity.

surveillance mechanism, i.e. the very asset of §3. It is compatible, however, with two other models. Under *credential-based* reputation, portable signed attestations are selectively disclosed: an agent proves “a verifier attests that I completed N tasks at quality q ” without revealing whom it transacted with, so trust travels with the agent rather than being reconstructed from observed traffic. Under *pairwise* reputation, two agents accrue trust over their own repeated interactions with no global observer. Both align with the verifiable-credentials direction already pursued in the ecosystem; only the global-ledger variant conflicts, and credential-based approaches can provide privacy-preserving alternatives while preserving the properties of §6. This is also the trust basis that the authentication mismatch of §8 requires.

10.2 Limitations

The properties of §6 are stated qualitatively; tightening them into adversary-indistinguishability games is future work. The strongest metadata protection (mixing, cover traffic) carries latency and bandwidth costs that may be unacceptable for interactive, low-latency agent calls; an identity-less transport also makes discovery and authentication harder, as §8 shows. Our evaluation (§9) measures inference and in-model actuation on simulated workflows anchored to a real capture; a reference binding with live wire measurements (latency, throughput, and an adversary’s reconstruction of G on real traffic), and a demonstration of

actuation against live agent traffic rather than within the model, are the natural next steps.

10.3 Deployment

Because A2A exposes custom protocol bindings, a metadata-protecting binding can be introduced incrementally and selected per Agent Card, coexisting with HTTP and SLIM bindings rather than replacing them. Agents for which the communication graph is sensitive (regulated, competitive, or adversarial settings) can opt in, while latency-sensitive agents retain existing bindings. The threat model and properties are transport- and protocol-agnostic, so the same analysis applies to MCP and to other transports on the frontier of §7.

11 Related Work

Threat modeling of agent-interop protocols.

Recent work has begun to systematize agent-interop security. Comparative threat models examine MCP, A2A, Agora, and ANP for protocol-specific and cross-protocol risks [3, 12], and surveys map the interoperability landscape [9]. These analyses concentrate on authentication, identity, message injection, permissioning, and the leakage of sensitive *payload* data: for instance, the sensitive context streamed during delegation [11]. Our surface is complementary and, to our knowledge, not previously treated as a first-class

transport-layer security surface: the transport-level *communication graph* (who communicates with whom, when), which persists even when payloads are fully protected.

Privacy leakage in multi-agent systems. A parallel line studies information leakage *within* multi-agent systems, showing that inter-agent channels leak substantially more than output channels and that privacy controls must extend to inter-agent communication [10, 16]. That work targets content-level leakage between agents; we target the metadata of the interactions themselves at the transport.

Anonymous communication. The properties we require (unlinkability, no central observer, metadata minimization) originate in the anonymous-communication literature: mix networks [6], onion routing [8], and modern mixnets such as Nym [13], surveyed broadly in [17]. Our contribution is not a new anonymity system but the application of these properties to agent-interop transport and an analysis of what an interop protocol must give up to obtain them (§8).

Inference and preemption from metadata. That metadata enables *semantic* inference is established for encrypted traffic by website-fingerprinting attacks [14]; that observable pending intent enables *preemption* is established by front-running and miner-extractable value in decentralized exchanges [7]. We argue (§5) that agent interoperability combines both, at machine speed, and our actuation result (§9.4) makes the bridge between them measurable: it casts the *value* of acting on recovered metadata as a decision-theoretic quantity (the advantage a budgeted adversary gains over a blind baseline), distinct from recoverability, and shows that a transport defense must drive that value down, not merely reduce recovery. We draw on these literatures for the inference and preemption primitives rather than extend them.

Credentials and unlinkability. The reconciliation of §10 draws on selectively disclosed verifiable credentials, whose unlinkable presentation is an active area; the W3C threat model for decentralized credentials catalogs the relevant attack surfaces [18]. These mechanisms supply trust without a global interaction graph, complementing the transport-level properties developed here.

12 Conclusion

The communication graph of interoperating agents remains exposed under today’s address-based bindings even with end-to-end payload encryption, and in agent systems it is more revealing than a privacy framing suggests. Because endpoints are often capability-labeled, workflows are structured, and interactions are action-coupled, the graph can leak *pending* workflows and hand an observer predictive leverage over machine-speed action. The exposure runs to the integrity and contestability of autonomous workflows, not their privacy alone. We gave a threat model for this surface, an account of what makes agent metadata distinctively revealing, transport- and bootstrap-layer properties against which any binding can be evaluated, an A2A case study in which pursuing those properties both surfaces and is constrained by the protocol’s implicit identity assumptions, and an empirical evaluation showing that the leakage is real and prospective, that this leakage carries decision-theoretic leverage (value to a budgeted adversary acting from a workflow’s opening), and that the properties, applied together, suppress both. A reference binding with live measurements, and a demonstration of actuation against live agent traffic rather than within the model, remain open.

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